

1 Дано:

$$R = 2 \text{ см} = 0,02 \text{ м}$$
$$\epsilon \approx 80$$

Решение:

$$C = \frac{Q}{\varphi} = \frac{Q}{\frac{1}{4\pi\epsilon_0\epsilon} \cdot \frac{Q}{R}} = 4\pi\epsilon_0\epsilon R =$$

$$= 4 \cdot 3,14 \cdot 8,85 \cdot 10^{-12} \cdot 80 \cdot 0,02 = 178 \cdot 10^{-12} \text{ ф}$$

Ответ: $178 \cdot 10^{-12} \text{ ф}$

2 Дано:

$$R_1 = 2 \text{ см} = 0,02 \text{ м}$$

$$R_2 = 6 \text{ см} = 0,06 \text{ м}$$

$$Q = 1 \text{ нКл} = 10^{-9} \text{ Кл}$$

$$\sigma_1 = ? \quad \sigma_2 = ?$$

Решение:

$$\varphi = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{R}$$

$$\sigma = \frac{Q}{S} = \frac{Q}{4\pi R^2}$$

$$\varphi_1 = \varphi_2 \Rightarrow \frac{1}{4\pi\epsilon_0} \cdot \frac{Q_1}{R_1} = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q_2}{R_2} \Rightarrow$$

$$Q_1 R_2 = Q_2 R_1$$

$$Q_1 + Q_2 = Q \Rightarrow Q_2 \cdot \frac{R_1}{R_2} + Q_2 = Q \Rightarrow$$

$$\Rightarrow Q_2 = \frac{Q}{\frac{R_1}{R_2} + 1} = \frac{10^{-9}}{\frac{0,02}{0,06} + 1} = 0,75 \cdot 10^{-9} \text{ Кл}$$

$$Q_1 = \frac{0,75 \cdot 10^{-9} \cdot 0,02}{0,06} = 0,25 \cdot 10^{-9} \text{ Кл}$$

$$\sigma_1 = \frac{Q_1}{4\pi R_1^2} = \frac{0,25 \cdot 10^{-9}}{4 \cdot 3,14 \cdot (0,02)^2} = 4,87 \cdot 10^{-9} \frac{\text{Кл}}{\text{м}^2}$$

$$\sigma_2 = \frac{Q_2}{4\pi R_2^2} = \frac{0,75 \cdot 10^{-9}}{4 \cdot 3,14 \cdot (0,06)^2} = 1,66 \cdot 10^{-9} \frac{\text{Кл}}{\text{м}^2}$$

3

Dano:

$$d = 1 \text{ cm} = 0,01 \text{ m}$$

$$\epsilon = 2$$

$$\Delta h = ?$$

Решение:

~~$$C = \frac{\epsilon_0 S}{h}$$~~

$$C_0 = \frac{\epsilon_0 S}{h}$$

$$C = \frac{S \epsilon_0}{(h' - d) + \frac{d}{\epsilon}}$$

$$\frac{1}{C_0} = \frac{1}{C} \Rightarrow h = (h' - d) + \frac{d}{\epsilon}$$

$$h' = h + d \left(1 - \frac{1}{\epsilon_r}\right)$$

$$\Delta h = h' - h = d \left(1 - \frac{1}{\epsilon_0}\right) = 1 \cdot \left(1 - \frac{1}{2}\right) = 0,5 \text{ cm}$$

$$\text{Отв: } 0,5 \text{ cm}$$

5 Dano:

$$R_1 = 10 \text{ cm} = 0,1 \text{ m}$$

$$R_2 = 10,2 \text{ cm} = 0,102 \text{ m}$$

$$Q = 5 \text{ nC} = 5 \cdot 10^{-9} \text{ C}$$

$$\epsilon = 2,1$$

$$U = ?$$

Решение:

~~$$U = \frac{Q}{4\pi\epsilon_0\epsilon} \left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$~~

$$E = \frac{Q}{4\pi\epsilon_0\epsilon r^2} \quad U = \int_{R_1}^{R_2} E dr$$

$$U = \frac{5 \cdot 10^{-9}}{4 \cdot 3,14 \cdot 8,85 \cdot 10^{-12} \cdot 2,1} \left(\frac{1}{0,1} - \frac{1}{0,102}\right) = 4138 \text{ B}$$

$$\text{Отв: } 4138 \text{ B}$$

4 Dano:

$$U_1 = 100 \text{ B}$$

$$\epsilon = 7$$

$$U_2 = ?$$

Решение:

$$C_1 = \frac{\epsilon \epsilon_0 S}{d}$$

$$C_2 = \frac{\epsilon_0 S}{d}$$

$$Q = C_1 U_1 = C_2 U_2$$

$$U_2 = \frac{C_1 U_1}{C_2} = \epsilon \cdot U_1 = 7 \cdot 100 = 700 \text{ B}$$

[6] Дано:

$$C_1 = 3 \text{ мкФ} = 3 \cdot 10^{-6} \text{ Ф}$$

$$C_2 = 6 \text{ мкФ} = 6 \cdot 10^{-6} \text{ Ф}$$

$$\varepsilon = 120 \text{ В}$$

$$Q_1 = ? \quad Q_2 = ?$$

$$U_1 = ? \quad U_2 = ?$$

$$1) \text{ --- } 2) \text{ --- }$$

Решение:

$$1) Q = UC \quad \varepsilon = U_1 = U_2 = 120 \text{ В}$$

$$Q_1 = U_1 C_1 = 120 \cdot 3 \cdot 10^{-6} = 360 \cdot 10^{-6} \text{ Кл}$$

$$Q_2 = U_2 C_2 = 120 \cdot 6 \cdot 10^{-6} = 720 \cdot 10^{-6} \text{ Кл}$$

$$2) \varepsilon = Q_1 + Q_2 \quad Q = Q_1 = Q_2 = \varepsilon C$$

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{C_1 + C_2}{C_1 \cdot C_2}$$

$$C = \frac{3 \cdot 10^{-6} \cdot 6 \cdot 10^{-6}}{(3+6) \cdot 10^{-6}} = 2 \cdot 10^{-6} \text{ Ф}$$

$$Q_1 = Q_2 = 120 \cdot 2 \cdot 10^{-6} = 240 \cdot 10^{-6} \text{ Кл}$$

$$U_1 = \frac{Q_1}{C_1} = \frac{240 \cdot 10^{-6}}{3 \cdot 10^{-6}} = 80 \text{ В}$$

$$U_2 = \frac{Q_2}{C_2} = \frac{240 \cdot 10^{-6}}{6 \cdot 10^{-6}} = 40 \text{ В}$$

[7] Дано:

$$d = 2 \text{ мм} = 0,002 \text{ м}$$

$$U = 1,8 \text{ кВ} = 1800 \text{ В}$$

$$\varepsilon = 7$$

$$k = ? \quad \sigma' = ?$$

Решение:

$$\varepsilon = 1 + \chi \quad \chi = \varepsilon - 1 = 7 - 1 = 6$$

$$P = \varepsilon_0 \chi E \quad \sigma' = P \quad E = \frac{U}{d} = \frac{1800}{0,002} = 900000 \text{ В/м}$$

$$\sigma' = 8,85 \cdot 10^{-12} \cdot 6 \cdot 900000 = 477,3 \cdot 10^{-7} \text{ Кл/м}^2$$

[8] Дано:

$$R = 5 \text{ см} = 0,05 \text{ м}$$

$$d = 2 \text{ мм} = 0,002 \text{ м}$$

$$Q = 10 \text{ нКл} = 10^{-8} \text{ Кл}$$

$$\varepsilon = 6$$

$$\sigma_1 = ? \quad \sigma_2 = ?$$

Решение:

$$E = \frac{1}{4\pi\varepsilon_0\varepsilon} \frac{Q}{R^2} \quad P = \varepsilon_0(\varepsilon - 1)E = \frac{(\varepsilon - 1)Q}{4\pi\varepsilon R^2}$$

$$\sigma_1 = -Pk = \frac{-(\varepsilon - 1)Q}{4\pi\varepsilon R^2} = \frac{-(6 - 1) \cdot 10^{-8}}{4 \cdot 3,14 \cdot 6 \cdot (0,05)^2} = -2,65 \cdot 10^{-7} \text{ Кл/м}^2$$

$$\sigma_2 = \frac{(\varepsilon - 1)Q}{4\pi\varepsilon(R+d)^2} = \frac{5 \cdot 10^{-8}}{4 \cdot 3,14 \cdot 6 \cdot (0,05 + 0,002)^2} = 1,36 \cdot 10^{-7} \text{ Кл/м}^2$$

[9] Дано:

$$E_0 = 2 \text{ МВ/м}$$

$$\varepsilon = 3$$

$$\sigma' = ?$$

Решение:

$$P = (\varepsilon - 1) \varepsilon_0 E$$

$$E = \frac{E_0}{\varepsilon}$$

$$\sigma' = P \Rightarrow \sigma = (\varepsilon - 1) \varepsilon_0 \frac{E_0}{\varepsilon} =$$

$$= (3 - 1) \cdot 8,85 \cdot 10^{-12} \cdot \frac{2 \cdot 10^6}{3} = 11,8 \cdot 10^{-6} \text{ Кл/м}^2$$